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Quantum Computing

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1

Complex linear algebra

1.1 Complex numbers

Complex number

A **Complex number** z is defined as:

$$z = a + ib$$

where $b \in \mathbb{R}$.

From a complex number we can define:

- Coniugate: $\bar{z} = \overline{a + ib} = a - ib$
- Modulus: $|z| = \sqrt{a^2 + b^2}$

Given a number z and its modulus $|z|$, we can find an angle $\theta \in [-\pi, \pi]$ such that:

$$\begin{aligned}\cos \theta &= \frac{a}{\sqrt{a^2 + b^2}} = \frac{a}{|z|} \\ \sin \theta &= \frac{b}{\sqrt{a^2 + b^2}} = \frac{b}{|z|}\end{aligned}$$

So, we can write:

$$z = |z|(\cos \theta + i \sin \theta)$$

We also define the exponential of an imaginary number:

$$e^{i\theta} = \cos \theta + i \sin \theta \implies z = |z|e^{i\theta}$$

1.2 Vectors and Matrices

A vector n is list of complex numbers, for which we can define a norm:

$$||v|| = \sqrt{|v_1|^2 + \dots + |v_n|^2} = \sqrt{\sum_{i=1}^n |v_i|^2}$$

Example:

Given the vector:

$$v = [4, i, 2 + 3i]$$

Its norm is:

$$||v|| = \sqrt{16 + 1 + 13} = \sqrt{30}$$

A matrix is a table of complex number, for which the multiplication is not commutative. The multiplication between a matrix A with size $m \times n$ and a matrix B with size $n \times q$ has size $m \times q$. The result matrix C has in every cell:

$$C_{i,j} = \sum_{k=0}^n A_{i,k} \cdot B_{k,j}$$

Example:

Given the matrices:

$$A = \begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$$

The multiplication $A \cdot B$ is:

$$C = \begin{bmatrix} 9 & 3 \\ 3+3i & 1+i \\ -12 & -4 \end{bmatrix}$$

1.2.1 Eigenvectors and Eigenvalues

A vector v is an **eigenvector** with **eigenvalue** λ for a matrix A if:

$$A \cdot v = \lambda \cdot v$$

We can also write:

$$A \cdot v - \lambda \cdot v = 0 \iff (A - \lambda I) \cdot v = 0$$

This equation has non-zero solutions only if:

$$\det(A - \lambda I) = 0$$

In general v and λ could be complex, but quantum mechanics uses matrices that always have eigenvalues $\lambda \in \mathbb{R}$.

2

Introduction

2.1 Probabilistic algorithms

In normal programming, an algorithms produce a fixed output for a given input. With probabilistic algorithms there is a finite set of possible outputs $\{o_1, \dots, o_N\}$ for a given input. Each outputs o_i has a probability of p_i to be produced, where:

$$\sum_{k=1}^N p_k = 1$$

Classical algorithms can have probabilistic outputs based on pseudo-random number generator (PRNG), that produce a sequence of numbers that appear random but after many elements the sequence repeats. In quantum algorithms there is no PRNG, but real randomness.

2.2 Qubit

Qubit

A **qubit** is the quantum equivalent of a bit, where the basis states are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The true state of a qubit ψ is a **superposition** defined as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ are called **probability amplitudes** and $|\alpha|^2 + |\beta|^2 = 1$.

The notation $|\psi\rangle$ is called "ket".

Example:

Given the vector:

$$v = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

The corresponding qubit is:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

The true state of a qubit cannot be observed, but only measured. Measuring a qubit makes it collapse into $|0\rangle$ or $|1\rangle$ with the probabilities $|\alpha|^2$ and $|\beta|^2$. We write this property as:

$$\alpha |0\rangle + \beta |1\rangle \begin{cases} |0\rangle & @ |\alpha|^2 \\ |1\rangle & @ |\beta|^2 \end{cases}$$

where @ is called "at".

For any complex number z such that $|z| = 1$, we have that $|z\alpha| = |\alpha|$, so when measuring we have that:

$$z\alpha |0\rangle + z\beta |1\rangle = \alpha |0\rangle + \beta |1\rangle$$

because the probability is based on the amplitudes squared.

2.2.1 Bloch sphere

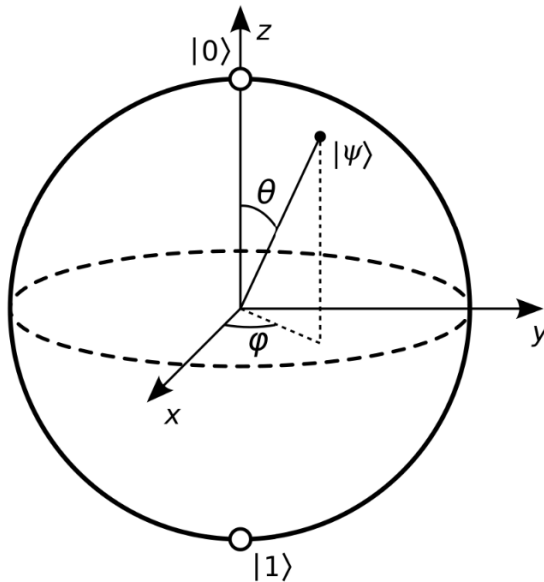
We can picture a qubit using the **Bloch sphere**. Considering a qubit:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

Rewriting it using polar coordinates:

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \right)$$

Since $|e^{i\gamma}| = 1$ we can remove it, representing the qubit based on two angles $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$.



There is no Bloch sphere equivalent for systems composed by two or more qubits.

2.3 Qubit operations

Qubits transformations are linear, so they can be represented as a matrix.

Pauli matrices

The **Pauli matrices** are 2×2 matrices that represent operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These are called the X, Y and Z gate respectively.

The result of applying this gates on the two basis states $|0\rangle$ and $|1\rangle$ are:

- X gate (also called NOT):

$$\sigma_x |0\rangle = |1\rangle \quad \sigma_x |1\rangle = |0\rangle$$

- Y gate:

$$\sigma_y |0\rangle = i |1\rangle \quad \sigma_y |1\rangle = -i |0\rangle$$

- Z gate:

$$\sigma_z |0\rangle = |0\rangle \quad \sigma_z |1\rangle = -|1\rangle$$

Another important transformation is the **Hadamard gate**, represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

This gate can map a classical state into a superposition:

$$H |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

If we measure it we will have a true random bits generator:

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \begin{cases} |0\rangle & @ \frac{1}{2} \\ |1\rangle & @ \frac{1}{2} \end{cases}$$

To be an allowed quantum transformation a matrix must preserve the norm of the input vector, so it must be **unitary**. Since $|\alpha|^2 + |\beta|^2 = 1$, the norm of every vector v must be $\|v\| = 1$ for any qubit state. An unitary matrix preserve the probabilities and is reversible, so from the output we can recover the input.

2.4 Multiple qubits

In classical computers we can describe n bits using an vector of elements and all the possible combination are obtained by performing the cartesian product:

$$\mathcal{B} = \{0, 1\} \implies \mathcal{B} \times \mathcal{B} = \{00, 01, 10, 11\}$$

The lenght of the vector grows linear in respect to n and each element is indipendent from the others.

To represent all the state of two qubits we need to use the **tensor product**. Given two qubits, the tensor product of their state vectors:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} \otimes \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0\beta_0 \\ \alpha_0\beta_1 \\ \alpha_1\beta_0 \\ \alpha_1\beta_1 \end{pmatrix} = \begin{pmatrix} \alpha_0 |\phi\rangle \\ \alpha_1 |\phi\rangle \end{pmatrix} = |\psi\phi\rangle$$

This product can also be written by using the **diagonal (or computational) basis vectors**, that are vectors with all 0 and a 1 in a single position i , called e_i .

$$\underbrace{\alpha_0\beta_0 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}}_{e_0} + \underbrace{\alpha_0\beta_1 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}}_{e_1} + \underbrace{\alpha_1\beta_0 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{e_2} + \underbrace{\alpha_1\beta_1 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{e_3} = \alpha_0\beta_0 |00\rangle + \alpha_0\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle$$

We can see that each vector e_i is equal to the vector state of $|i\rangle$ in which we consider the binary representation of i . So, the previous result can also be written as a sum:

$$\sum_{i,j \in \{0,1\}} \alpha_i \beta_j |ij\rangle$$

Example:

Given two qubits:

$$|\psi\rangle = |\phi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

Their tensor product is:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \otimes \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

2.4.1 Entanglement

In the example above the two qubits can be separated and decomposed into two valid qubit states.

Since any qubit written:

$$\alpha_0 |00\rangle + \alpha_1 |01\rangle + \alpha_2 |10\rangle + \alpha_3 |11\rangle \quad \text{with} \quad \sum_i |\alpha_i|^2 = 1$$

is a valid quantum state for a 2-qubit system we can have a valid qubit such as:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

It's not possible to separate this state in two different quantum states for which their tensor product is $|\psi\rangle$. This would require:

$$|a\rangle \otimes |b\rangle = \alpha_0\beta_0 |00\rangle + \alpha_0\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle = \begin{cases} \alpha_0\beta_0 = \alpha_1\beta_1 = 0 \\ \alpha_0\beta_1 = \alpha_1\beta_0 = \frac{1}{\sqrt{2}} \end{cases}$$

The product $\alpha_0\beta_0 = 0$ implies that at least one of the two is 0, but this means also that at least one of $\alpha_0\beta_1$ or $\alpha_1\beta_0$ is 0.

This state is **entangled**, it means that it doesn't describe two qubit states but the entire 2-qubit system as a whole. This entangled states are called the **EPR** states and when measuring them we see that:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \begin{cases} |01\rangle & @ \frac{1}{2} \\ |10\rangle & @ \frac{1}{2} \end{cases}$$

This pairs also have a strange behaviour when physically separated. Since we cannot measure $|00\rangle$ or $|11\rangle$, measuring one of the two qubit is enough to know the entire system, also when the two qubits are kilometers apart. In fact is proved that if we take the two qubits apart and measure one of the two, the other will also collapse its state.

2.4.2 Mutiple qubits gates

The basic gate for two qubits its the CNOT gate, where the first bit is the **control** and the second one the **target**. The target bit is flipped when the control is 1, so the truth table is:

a	b	CNOT(a,b)
0	0	0
0	1	1
1	0	1
1	1	0

Logically we have that:

$$\text{CNOT}(a, b) = a \oplus b$$

Since we want the computation to be invertible we need the value of a to recover the value of b , so the quantum CNOT behaves as follows:

$$\text{CNOT}(|a\rangle \otimes |b\rangle) = |a\rangle \otimes |a \oplus b\rangle = |a, a \oplus b\rangle$$

The matrix that represent this gate is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

2.5 Quantum circuit

In quantum computing we program using gates and circuits. Each circuit U must be unitary, that means:

$$UU^\dagger = U^\dagger U = I$$

Where $U^\dagger$ is the **adjoint** of U , given by $U_{i,j}^\dagger = U_{i,j}^*$.

The notation used for the gates seen before is:

- X, Y, Z and H gates: single bit with the correspondig letter

$$|a\rangle \text{ --- } \boxed{X/Y/Z/H} \text{ --- } |\bar{a}\rangle$$

- Measurment gate: the double wire after means that is passing classical bits, not quantum ones

$$\alpha|0\rangle + \beta|1\rangle \text{ --- } \boxed{\text{Measurement}} \text{ --- } \begin{cases} |0\rangle & @ |\alpha|^2 \\ |1\rangle & @ |\beta|^2 \end{cases}$$

- CNOT gate: the wire with the dot is the control and the one with the gato or xor is the target

$$\begin{array}{c} |a\rangle \text{ --- } \bullet \text{ --- } |a\rangle \\ |b\rangle \text{ --- } \oplus \text{ --- } |a \oplus b\rangle \end{array}$$

$$\begin{array}{c} |a\rangle \text{ --- } \bullet \text{ --- } |a\rangle \\ |b\rangle \text{ --- } \boxed{X} \text{ --- } |a \oplus b\rangle \end{array}$$

2.5.1 Creating EPR pairs

There are different circuits capable of creating EPR pairs, one of the common is composed by H and CNOT gates.

$$\begin{array}{c} |0\rangle \text{ --- } \boxed{H} \text{ --- } \bullet \text{ --- } \\ |1\rangle \text{ --- } \text{ --- } \boxed{X} \text{ --- } \end{array} \left. \vphantom{\begin{array}{c} |0\rangle \\ |1\rangle \end{array}} \right\} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

We can evaluate the circuit by sequentially crossing the gates from left to right:

$$|0\rangle \otimes |1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

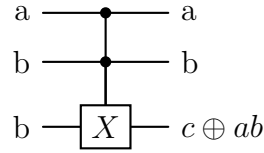
Another gate that creates EPR pairs, but with 3 qubits is the GHZ (Greenberger, Horne, Zeilinger) gate:

$$\begin{array}{c} |0\rangle \text{ --- } \boxed{H} \text{ --- } \bullet \text{ --- } \bullet \text{ --- } \\ |0\rangle \text{ --- } \text{ --- } \boxed{X} \text{ --- } \text{ --- } \\ |0\rangle \text{ --- } \text{ --- } \text{ --- } \boxed{X} \text{ --- } \end{array} \left. \vphantom{\begin{array}{c} |0\rangle \\ |0\rangle \\ |0\rangle \end{array}} \right\} \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

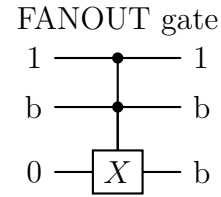
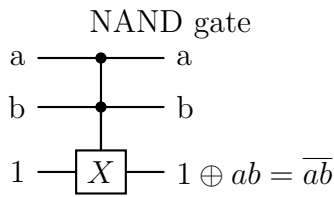
Tecnically all quantum circuits starts with all $|0\rangle$ as inputs, but we can simply add a NOT gate to change it to $|1\rangle$.

2.5.2 Reversible classical computing

Using a specific quantum gate, called the **Toffoli** gate (also called CCNOT), we make classical computing reversible.



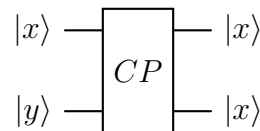
With this gate we can create the NAND and FANOUT gate, that means that any boolean circuit can be implemented only with Toffoli gate, so its reversible.



2.6 Cloning and Teleportation

2.6.1 No cloning theorem

A copy operation takes a source variable to be copied and a destination variable where the copy is applied. As a circuit we could write:



We need a variable $|y\rangle$ that is a "white paper" where the content of $|x\rangle$ can be copied, but this cannot exist.

No cloning theorem

There is no operation CP, such that:

$$\exists |y\rangle \forall |x\rangle \text{ CP}(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |x\rangle$$

Proof by contradiction:

For the proof the $|\rangle$ notation will be avoided.

CP must work on superposed states, so:

$$\exists y \forall x, a \text{ CP}(\underbrace{(x+a) \otimes y}_{(x \otimes y) + (a \otimes y)}) = (x+a) \otimes (x+a)$$

CP must be linear:

$$\exists y \forall x, a \text{ CP}(x \otimes y) + \text{CP}(a \otimes y) = (x + a) \otimes (x + a)$$

Since CP copies its input:

$$\forall x, a \text{ } (x \otimes x) + (a \otimes a) = (x + a) \otimes (x + a) = (x \otimes x) + (x \otimes a) + (a \otimes x) + (a \otimes a)$$

But we can prove with a simple example like $x = |0\rangle$ and $a = |1\rangle$ that this equation is not true.

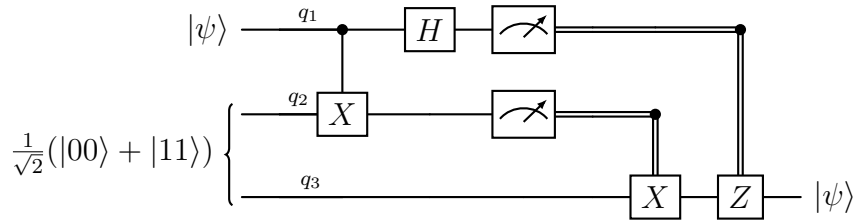
$$\left. \begin{array}{l} x = |0\rangle \\ a = |1\rangle \end{array} \right\} \Rightarrow (x \otimes x) + (a \otimes a) = |00\rangle + |11\rangle \neq (x \otimes x) + (x \otimes a) + (a \otimes x) + (a \otimes a) = |00\rangle + |01\rangle + |10\rangle + |11\rangle$$

This means that we cannot clone perfectly, but **imperfect cloning** with high probability is possible.

2.6.2 Quantum teleportation

Even if we cannot perfectly clone a qubit, we can teleport it perfectly fine.

Suppose Alice has a qubit $|\psi\rangle$ and she wants to send it to Bob, this can be achieved by using 2 qubits and 2 classical bits. The Quantum teleportation circuit is the following:



We can give a pseudocode for the algorithm:

Algorithm: Quantum Teleportation

```
def teleport(|psi>):
    q1, q2, q3 = |psi> 1/sqrt(2) * (|00> + |11>)
    q1, q2 = CNOT(q1, q2)
    q1 = H(q1)
    b1, b2 = Measure(q1, q2)
    b2, q3 = CNOT(b2, q3)
    b1, q3 = CZ(b1, q3)
    return q3
```

We can prove that the algorithm works by executing the operations done by the code step by step, starting from a random qubit ψ .

$$|\psi\rangle \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (\alpha|0\rangle + \beta|1\rangle) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} [\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)]$$

Applying the CNOT on q_1 and q_2 we obtain:

$$\frac{1}{\sqrt{2}} [\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)]$$

Next, we apply the Hadamard gate on q_1 :

$$\begin{aligned} & \frac{1}{\sqrt{2}} \left[\alpha \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)(|10\rangle + |01\rangle) \right] = \\ & \frac{1}{2} [\alpha(|000\rangle + |011\rangle + |100\rangle + |111\rangle) + \beta(|010\rangle + |001\rangle - |110\rangle - |101\rangle)] = \\ & \frac{1}{2} [|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\alpha|1\rangle + \beta|0\rangle) + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\alpha|1\rangle - \beta|0\rangle)] \end{aligned}$$

Now we have four possible outcomes based on the measurement done by Alice. Given the measured bits Bob need to apply different operation on its remaining qubit, to obtain the initial qubit $|\psi\rangle$ ($\alpha|0\rangle + \beta|1\rangle$).

Alice bits	Bob qubit	Operations
00	$ \phi\rangle = \alpha 0\rangle + \beta 1\rangle$	$I \phi\rangle = \psi\rangle$
01	$ \phi\rangle = \alpha 1\rangle + \beta 0\rangle$	$X \phi\rangle = \psi\rangle$
10	$ \phi\rangle = \alpha 0\rangle - \beta 1\rangle$	$Z \phi\rangle = \psi\rangle$
11	$ \phi\rangle = \alpha 1\rangle - \beta 0\rangle$	$XZ \phi\rangle = \psi\rangle$

Since Alice has measured her qubit, the qubit collapsed and she lost it forever, so now only Bob has that qubit.

2.7 Quantum parallelism

Given any boolean function $f : \mathcal{B}^n \rightarrow \mathcal{B}$, we can show that the function:

$$U_f = (x, y) \in (\mathcal{B}^n, \mathcal{B}) \rightarrow (x, y \oplus f(x))$$

Is reversible and unitary, so its a valid quantum gate.

Example:

For a 1-bit function we have the truth table:

x	y	$y \oplus f(x)$	
		$f(x) = 0$	$f(x) = 1$
0	0	0	1
0	1	0	0
1	0	1	1
1	1	1	0

So, we have that:

$$\begin{cases} U_f(x, 0) = (x, f(x)) \\ U_f(x, 1) = (x, \overline{f(x)}) \end{cases}$$

Since the function U_f can be implemented unitarily, we can apply it to qubits, obtaining:

$$\begin{cases} U_f|x \otimes 0\rangle = |x \otimes f(x)\rangle \\ U_f|x \otimes 1\rangle = |x \otimes \overline{f(x)}\rangle \end{cases}$$

Example:

Given $|x\rangle, |y\rangle$ as:

$$|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad |y\rangle = |0\rangle$$

We can apply U_f to obtain:

$$U_f \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \right) = \frac{1}{\sqrt{2}}(U_f |00\rangle + U_f |10\rangle) = \frac{1}{\sqrt{2}}(|0 \otimes f(0)\rangle + |1 \otimes f(1)\rangle)$$

So $f(0)$ and $f(1)$ are computed in **parallel**.